MATHS PBL

TOPIC: Linear Differential Equations in Financial Modelling

Project Goal: To model various financial scenarios, including investment growth, continuous savings, and debt amortization, using linear differential equations and their analytical solutions.

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SUBTOPICS:

1. The Foundation - Discrete vs Continuous Growth
2. Solving the Basic Growth Equation
3. Modelling Continuous Investment (Non-Homogeneous)
4. Modelling Debt and Retirement (Negative Flow)
5. Comparison and Real-World Limitations

The Foundation - Discrete vs. Continuous Growth

1. Introduction to Interest and Compounding

The project begins by distinguishing between **simple interest** (linear growth) and **compound interest** (exponential growth). The key focus is on the power of compounding and how it relates to the frequency (n times per year). The general discrete compounding formula is:

$$A(t) = P\left(1 + \frac{r}{n}\right)^{nt}$$

2. The Shift to Continuous Compounding

Show the mathematical transition as the compounding frequency approaches infinity

(n approaches to infinity).

$$A(t) = \lim_{n \to \infty} P\left(1 + \frac{r}{n}\right)^{nt} = Pe^{rt}$$

This formula, A(t) = P e^rt, represents the maximum possible growth rate for a given nominal rate r and serves as the solution to the simplest LDE model.

3. Deriving the Fundamental Linear Differential Equation (LDE)

The continuous model implies that the rate of change of money, dA/dt, is directly proportional to the amount of money currently present, A. This is because more money earns more interest instantly.

The fundamental first-order homogeneous LDE is derived as:

$$\frac{dA}{dt} = rA$$

where r is the continuous interest rate. This equation states that the rate of change of the amount A is proportional to A.

WORKING EXAMPLE:

Compounding Frequency (n)	Formula Used	Calculation	Future Value (A) After 10 Years
Annually ($n = 1$)	$A = 5000\left(1 + \frac{0.06}{1}\right)^{(1\times10)}$	$A = 5000(1.06)^{10}$	**$8,954.24**
Monthly ($n = 12$)	$A = 5000\left(1 + \frac{0.06}{12}\right)^{(12\times10)}$	$A = 5000(1.005)^{120}$	**$9,096.98**
Continuously ($n \to \infty$)	$A = 5000e^{(0.06\times10)}$	$A = 5000e^{0.6}$	**$9,110.59**

4. Required Deliverables

- A mathematical explanation of the limit process leading to P e^rt.
- A clear derivation of the LDE: dA/dt = rA.

- A working example comparison table showing the future value of a \$5,000 investment after 10 years at 6% interest compounded annually, monthly, and continuously.

Solving the Basic Growth Equation

1. Analytical Solution using Separation of Variables

This module focuses on solving the basic LDE derived by Student 1. The equation is

$$\frac{dA}{dt} = rA.$$

Using separation of variables:

$$\frac{dA}{A} = r\,dt$$

Integrating both sides:

$$\int \frac{1}{A}\,dA = \int r\,dt$$

$$\ln|A| = rt + C$$

Exponentiating both sides leads to the general solution:

$$A(t) = e^{rt+C} = e^{C}e^{rt}.$$

2. The Initial Value Problem (IVP)

The constant of integration C is determined by the initial condition,

A(0) = A_0 (the initial principal).

$$A_0 = e^C e^{r(0)} = e^C$$

Substituting A_0 for e^C yields the final, specific solution:

$$A(t) = A_0 e^{rt}$$

3. Doubling Time Analysis

A key application is finding the time T_D it takes for an investment to double

(A(T_D) = 2A_0).

$$2A_0 = A_0 e^{rT_D} \implies 2 = e^{rT_D}$$

$$T_D = \frac{\ln(2)}{r}$$

This solution is the basis for the **"Rule of 72,"** a quick financial approximation.

WORKING EXAMPLE:

Set up the equation using the tripling condition:

$$3A_0 = A_0 e^{rt}$$

Cancel the initial principal (A_0):

$$3 = e^{rt}$$

Take the natural logarithm ($\ln$) of both sides to isolate the exponent:

$$\ln(3) = \ln(e^{rt})$$

$$\ln(3) = rt$$

Solve for t:

$$t = \frac{\ln(3)}{r}$$

Substitute the given rate ($r = 0.08$):

$$t = \frac{\ln(3)}{0.08}$$

Calculate the numerical value:

$$t \approx \frac{1.0986}{0.08}$$

$$\mathbf{t} \approx \mathbf{13.73} \text{ years}$$

Conclusion: It takes approximately **13.73 years** for the investment to triple at an 8% continuous rate.

4. Required Deliverables

- A step-by-step analytical solution of dA/dt = rA using **separation of variables**.
- A clear derivation of the doubling time formula, T_D = ln(2)/r.
- A worked example calculating the time required to triple an investment at an 8% continuous rate.

Modelling Continuous Investment (Non-Homogeneous)

1. Formulating the Non-Homogeneous LDE

Introduce the scenario where an investor has an initial principal A_0 and makes **continuous deposits** at a constant annual rate D. This adds an external, non-homogeneous term to the LDE:

$$\frac{dA}{dt} = rA + D$$

Rewritten in standard linear form:

$$\frac{dA}{dt} - rA = D$$

2. Solution using the Integrating Factor Method

The integrating factor (IF)

$$\mu(t) = e^{\int P(t)\, dt}$$

is used, where P(t) = -r:

$$\mu(t) = e^{\int -r\, dt} = e^{-rt}$$

Multiply the standard equation by mu(t), recognizing that the left side becomes the derivative of a product:

$$\frac{d}{dt}(Ae^{-rt}) = De^{-rt}$$

Integrating and solving for A(t) gives the general solution:

$$A(t) = \left(A_0 + \frac{D}{r}\right) e^{rt} - \frac{D}{r}$$

3. Interpretation and Future Value of an Annuity

Interpret the two components of the solution: A_0 e^rt (growth of the initial lump sum) and (D/r)(e^{rt- 1}) (growth due to the continuous deposits, or the **Future Value of the Continuous Annuity**).

WORKING EXAMPLE:

Goal: Calculate $A(20)$ for $A_0 = 1000$, $D = 5000$, $r = 0.07$.

Formula:

$$A(t) = \left(A_0 + \frac{D}{r}\right) e^{rt} - \frac{D}{r}$$

Calculation:

1. **Substitute Values:**

$$A(20) = \left(1000 + \frac{5000}{0.07}\right) e^{(0.07 \times 20)} - \frac{5000}{0.07}$$

2. **Evaluate $\frac{D}{r}$ and e^{rt}:**

$$\frac{5000}{0.07} \approx 71,428.57$$

$$e^{1.4} \approx 4.0552$$

3. **Final Result:**

$$A(20) \approx (1000 + 71,428.57) \times 4.0552 - 71,428.57$$

$$A(20) \approx 293,491.54 - 71,428.57$$

$$\mathbf{A(20) \approx \$222,062.97}$$

4. Required Deliverables

- A clear explanation of why continuous deposits create a **non-homogeneous** equation.
- A step-by-step solution of dA/dt - rA = D using the **Integrating Factor Method**.
- A worked example showing the value of a retirement account after 20 years, starting with $1,000 and continuously depositing $5,000 per year at a 7% rate.

Modelling Debt and Retirement (Negative Flow)

1. The Amortization Model

This module applies the non-homogeneous LDE to scenarios where money is continuously **withdrawn** or paid out (loan amortization or retirement drawdown). The LDE changes only in the sign of the constant external flow W:

$$\frac{dA}{dt} = rA - W$$

Rewritten in standard linear form:

$$\frac{dA}{dt} - rA = -W$$

2. Loan Amortization: Solving for the Payment

The goal is to find the payment rate W needed to reduce the initial loan A_0 to zero

(A(T) = 0) over a term T.

The solution for the balance A(t) is:

$$A(t) = \left(A_0 - \frac{W}{r}\right) e^{rt} + \frac{W}{r}$$

Setting A(T) = 0 and solving for W yields the required payment formula:

$$W = \frac{rA_0}{1 - e^{-rT}}$$

3. Retirement Withdrawal: Sustainable Income

Model a retirement account A_0 used to fund continuous withdrawals W. Show how to calculate W given a term T. Furthermore, calculate the **perpetual withdrawal rate** dA/dt=0 where the interest earned exactly equals the withdrawal amount: rA - W = 0 therefore W = rA.

WORKING EXAMPLE:

Set $A(t) = 0$ at $t = T$:

$$0 = \left(A_0 - \frac{W}{r}\right) e^{rT} + \frac{W}{r}$$

Solve for W:

$$W = \frac{rA_0e^{rT}}{e^{rT} - 1}$$

Substitute $A_0 = 150000$, $r = 0.04$, and $T = 20$:

Final Result:

$$W = \frac{13,353.24}{1.22554} \approx \mathbf{\$10,896.25} \text{ per year}$$

conclusion

The required continuous annual payment rate to pay off the $150,000 loan in 20 years at a 4% continuous interest rate is approximately **$10,896.25 per year**.

4. Required Deliverables

- The step-by-step derivation of the payment formula W = {r A_0}/{1 - e^{-rT}.
- A worked example showing the continuous payment rate required to pay off a $150,000, 20-year loan at 4% interest.
- A discussion on the concept of sustainable income from an investment.

Comparison and Real-World Limitations

Summary of LDE Solutions

Summarize the three core financial LDE models and their solutions:

Scenario	Differential Equation	Analytical Solution
Simple Growth	$\frac{dA}{dt} = rA$	$A(t) = A_0 e^{rt}$
Investment/Savings	$\frac{dA}{dt} = rA + D$	$A(t) = \left(A_0 + \frac{D}{r}\right) e^{rt} - \frac{D}{r}$
Amortization/Debt	$\frac{dA}{dt} = rA - W$	$A(t) = \left(A_0 - \frac{W}{r}\right) e^{rt} + \frac{W}{r}$

2. Limitations of the Continuous Model

Critically analyze the shortcomings of using constant-rate LDEs in real finance:

- **Discrete vs. Continuous:** Actual financial transactions (deposits, payments) occur monthly or annually, not continuously. The LDE is an approximation.
- **Variable Rates:** Interest rates (r) and inflation rates (i) are not constant over time. This would require solving a more complex LDE with a **variable coefficient** $r(t)$.
- **Taxes and Fees:** The models do not account for taxes or transaction fees, which reduce the effective growth rate.

3. Introduction to Dynamic Financial Modelling

Briefly introduce how advanced financial modeling moves beyond simple LDEs:

- **Stochastic Processes:** Used to model random market fluctuations (e.g., Brownian motion, leading to the **Black-Scholes equation**, a partial differential equation).
- **Non-Linear Models:** Mention how market saturation or resource limits could require **non-linear differential equations** (e.g., the Logistic model for market share growth).

4. Required Deliverables

- A table summarizing the three main LDEs and their solutions (as shown above).
- A critical discussion of the limitations of using a constant-coefficient LDE in a dynamic economy.
- A final conclusion summarizing the power and scope of differential equations in modern quantitative finance.